



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

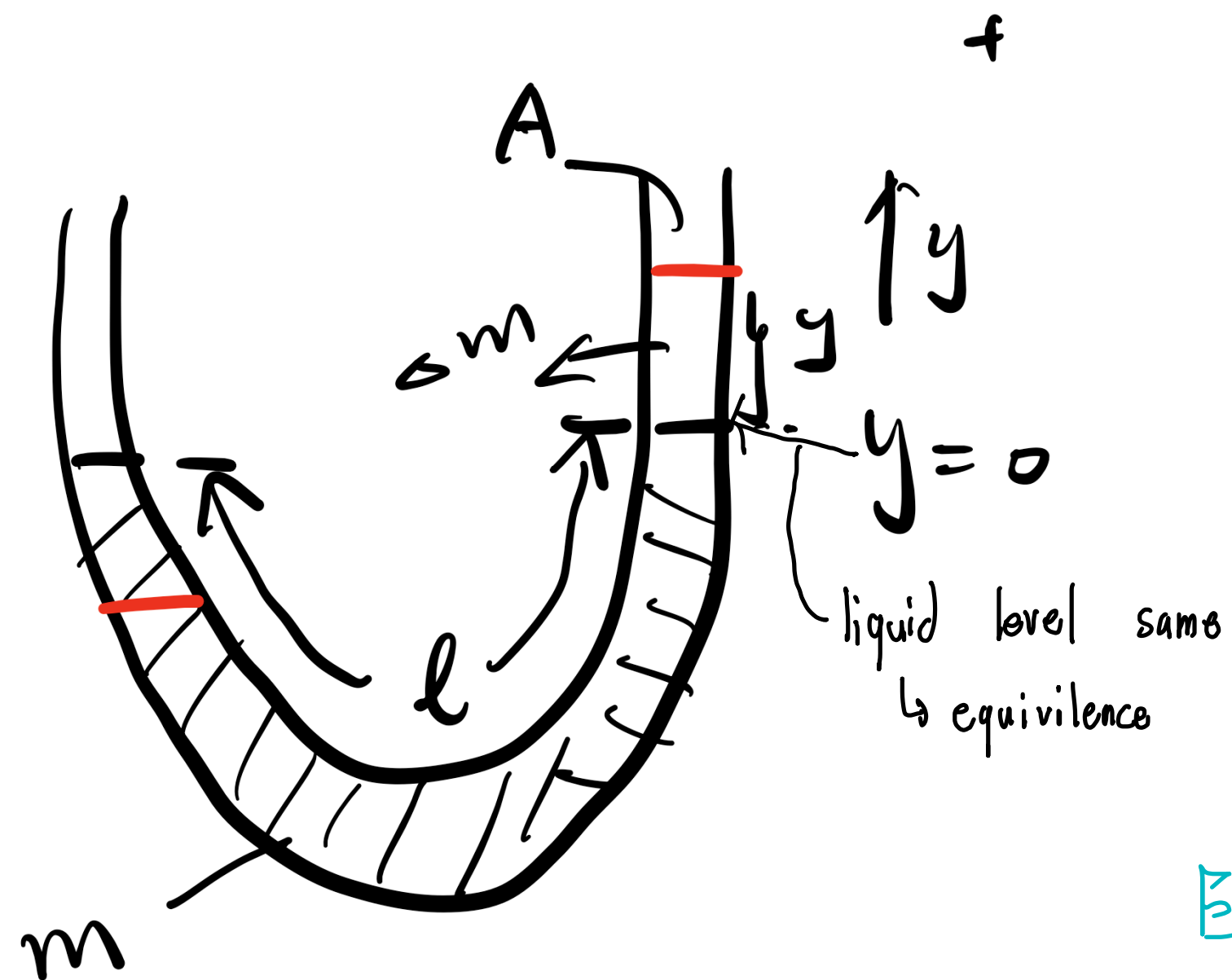
Lecture 23: Oscillation

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Recap: Oscillation

- Simple harmonic motion, Equation of motion: $\frac{d^2x}{dt^2} + Bx = 0$
- Spring-object system
 - Equation of Motion: $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
 - Solution: $x = A \cos(\omega t + \phi)$, where
 - Angular frequency: $\omega = \sqrt{\frac{k}{m}}$
 - Period: $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$
 - Frequency: $f = \frac{1}{T}$
- Simple pendulum
 - $\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$
 - $\theta = A \cos(\omega t + \phi)$, where
 - $\omega = \sqrt{\frac{g}{l}}$
 - $T = 2\pi\sqrt{\frac{l}{g}}$
 - $f = \frac{1}{T}$

Fluid & Oscillation



EOM

SHM

$$F = 2 \cdot \Delta m \cdot g = -2 \frac{P \cdot y \cdot A \cdot g}{g}$$

$$= m a = \frac{P \cdot l \cdot A \cdot g}{g}$$

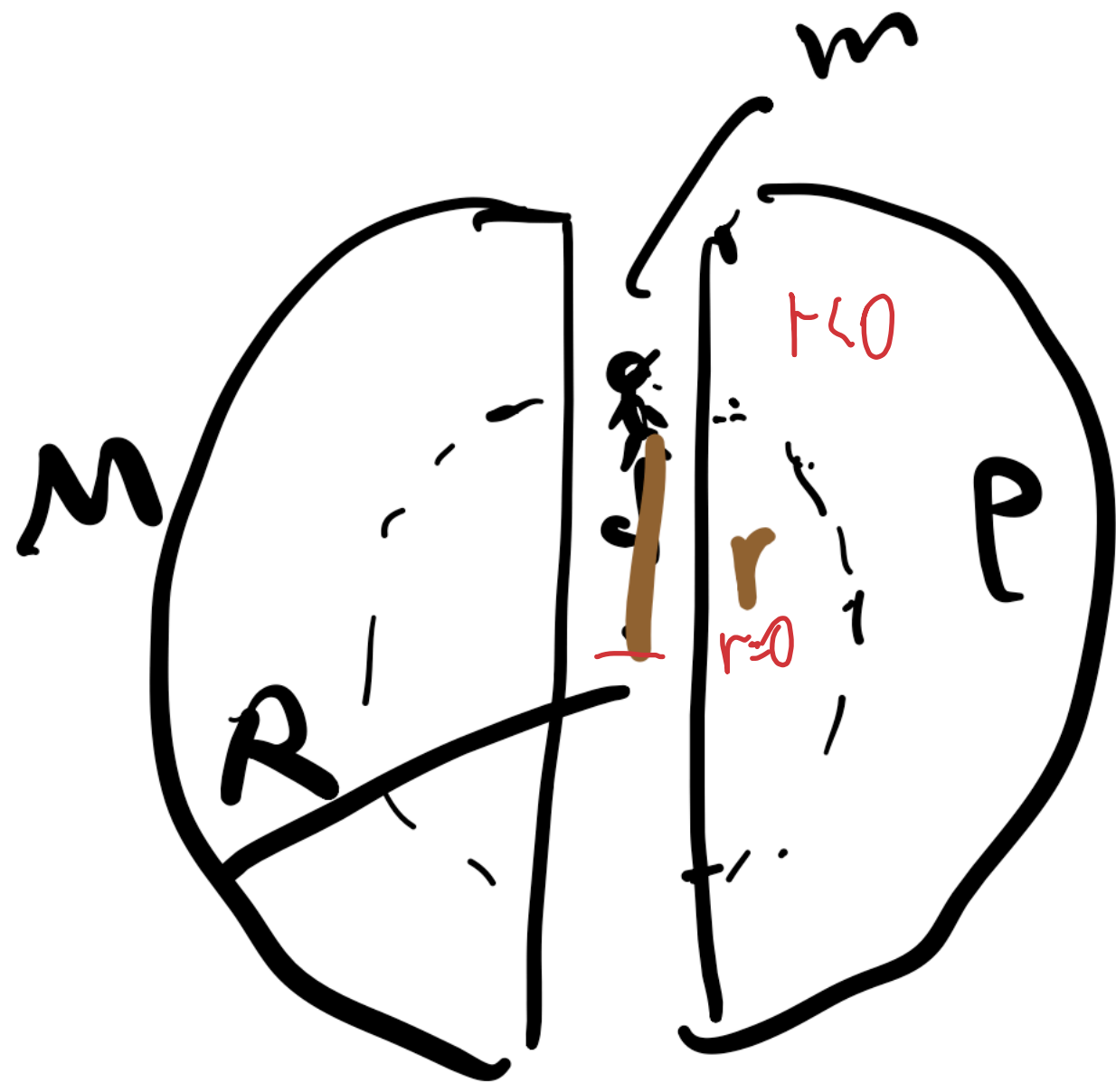
$$l a + 2 g y = 0$$

$$l \frac{d^2 y}{dt^2} + 2 g y = 0$$

$$\frac{d^2 y}{dt^2} + \frac{2 g}{l} y = 0$$

$$\omega = \sqrt{\frac{2 g}{l}}$$

Gravitation & Oscillation



$$F_g = -G \cdot \frac{\rho \cdot \frac{4\pi}{3} r^3 \cdot m}{r^2}$$

$$= -G \rho \frac{4\pi}{3} m \cdot r$$

$$= m a$$

$$= m \frac{d^2 r}{dt^2}$$

$$\text{SHM} \quad \frac{d^2 r}{dt^2} + G \rho \frac{4\pi}{3} r = 0$$

$$\omega = \sqrt{G \rho \frac{4\pi}{3}}$$

$$\frac{4\pi}{3} \rho R^3 = M$$

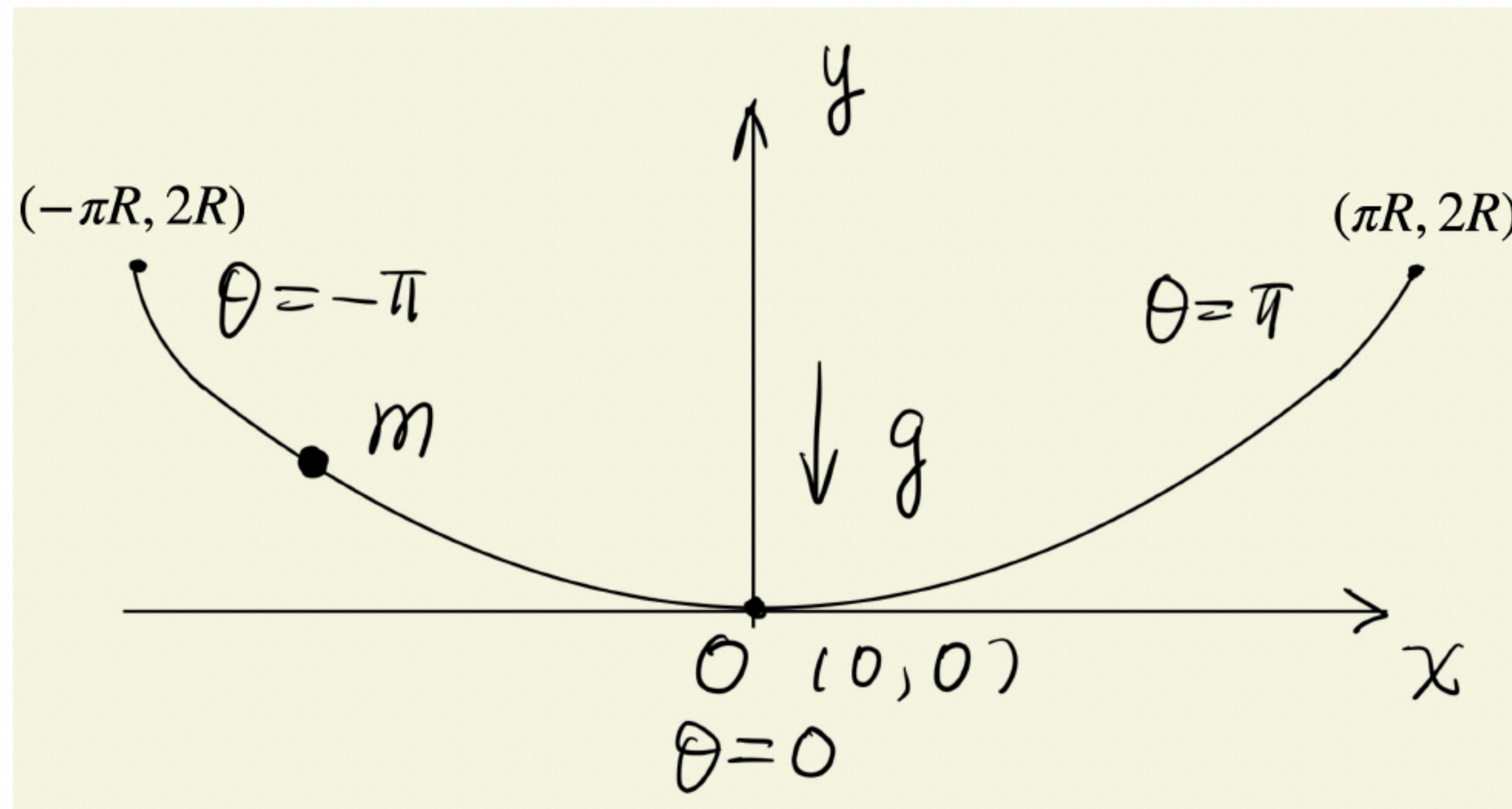
11. Oscillation on the Cycloidal Wire.

(10 pts)

A bead of mass m moves on a smooth (frictionless) wire shaped as a cycloid:

$$x = R(\theta + \sin \theta), \quad y = R(1 - \cos \theta),$$

where R is a positive constant and the parameter θ lies in the range $-\pi \leq \theta \leq \pi$.



The wire is fixed at point O , with the y -axis pointing vertically upwards and the x -axis pointing horizontally. As shown in the figure above, the gravitational force acts in the $-y$ direction. Since the cycloidal wire can be parametrized by θ , one can describe the motion of the bead on the wire in terms of $\theta(t)$, which is a function of time t .

(a) Find the x and y components of the velocity of the bead. **(2 pts)**

Hint: $v_x = \frac{dx}{dt}$ and $v_y = \frac{dy}{dt}$. Express your results in term of θ and the time derivative of θ ,
which is given as $\dot{\theta} \equiv \frac{d\theta}{dt}$. $x = R(\theta + \sin \theta), \quad y = R(1 - \cos \theta),$

$$v_x = R \dot{\theta} + R \frac{d \sin \theta}{d\theta} = R \dot{\theta} + R \cos \theta \cdot \dot{\theta} = R \dot{\theta} (1 + \cos \theta)$$

$$v_y = R \sin \theta \cdot \dot{\theta}$$

(b) Find the kinetic energy of the bead sliding on the wire. **(1 pt)**

$$K = \frac{1}{2} m (v_x^2 + v_y^2) = m R^2 \dot{\theta}^2 (1 + \cos \theta)$$

(c) Consider the gravitational potential together with the kinetic energy, and show that the mechanical energy of the bead sliding on the wire

$$E = mR^2 \dot{\theta}^2 [1 + \cos \theta] + mgR(1 - \cos \theta)$$

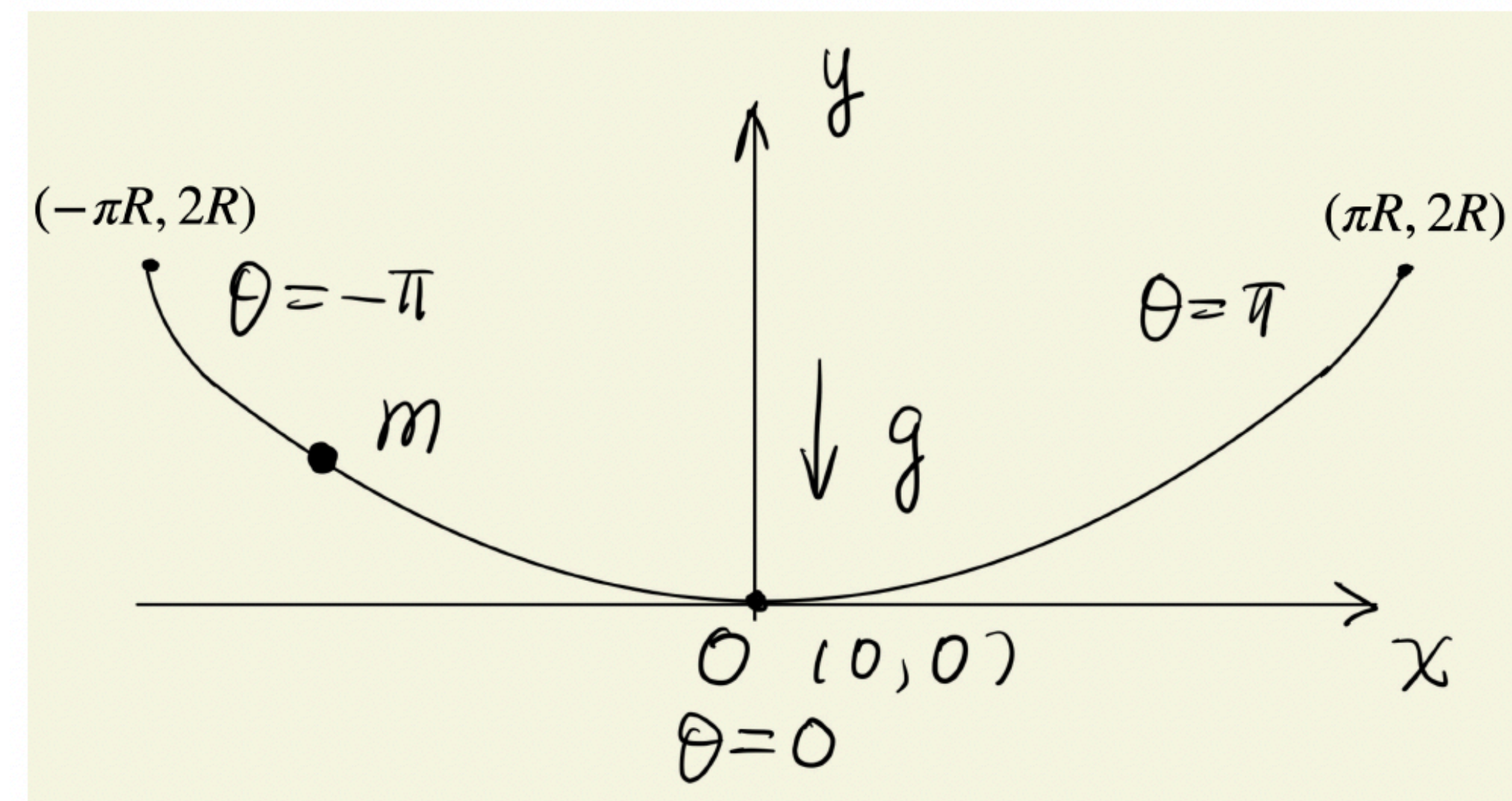
is a constant.

(1 pt)

$$E = K + U$$

$$U = mgR(1 - \cos \theta)$$

$$E = mR^2 \dot{\theta}^2 (1 + \cos \theta) + mgR(1 - \cos \theta)$$



(d) By changing variable from θ to $u \equiv \sin \frac{\theta}{2}$, rewrite the expression for E from Part (c). Hint:

Express your results in terms of the new variables u and $\dot{u} \equiv \frac{du}{dt}$.

(2 pts)

You may find these two trigonometric identities useful:

$$\sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2} \quad \text{and} \quad \cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}.$$

$$E = mR^2 \dot{\theta}^2 (1 + \cos \theta) + mgk(1 - \cos \theta)$$

$$\dot{u} = \frac{1}{2} \cos \frac{\theta}{2} \cdot \dot{\theta}$$

$$= mR^2 \dot{\theta}^2 \left(2 \cos^2 \frac{\theta}{2} \right) + mgk(2u^2)$$

$$= 8mR^2 \dot{u}^2 + 2mgku^2$$

- (e) By noting that E is a constant and differentiating $E(u, \dot{u})$ with respect to t , derive the equation of motion for the bead. **(2 pts)**

$$\frac{dE}{dt} = 16mR^2 \dot{u} \cdot \ddot{u} + 4mgRu \cdot \ddot{u} = 0$$

$$4R\ddot{u} + gu = 0$$

$$\frac{d^2u}{dt^2} + \frac{g}{4R}u = 0$$

- (f) Based on the above equation of motion, is the oscillation of the bead about $x = 0$ a simple harmonic motion? (Yes/No) **(1 pt)**

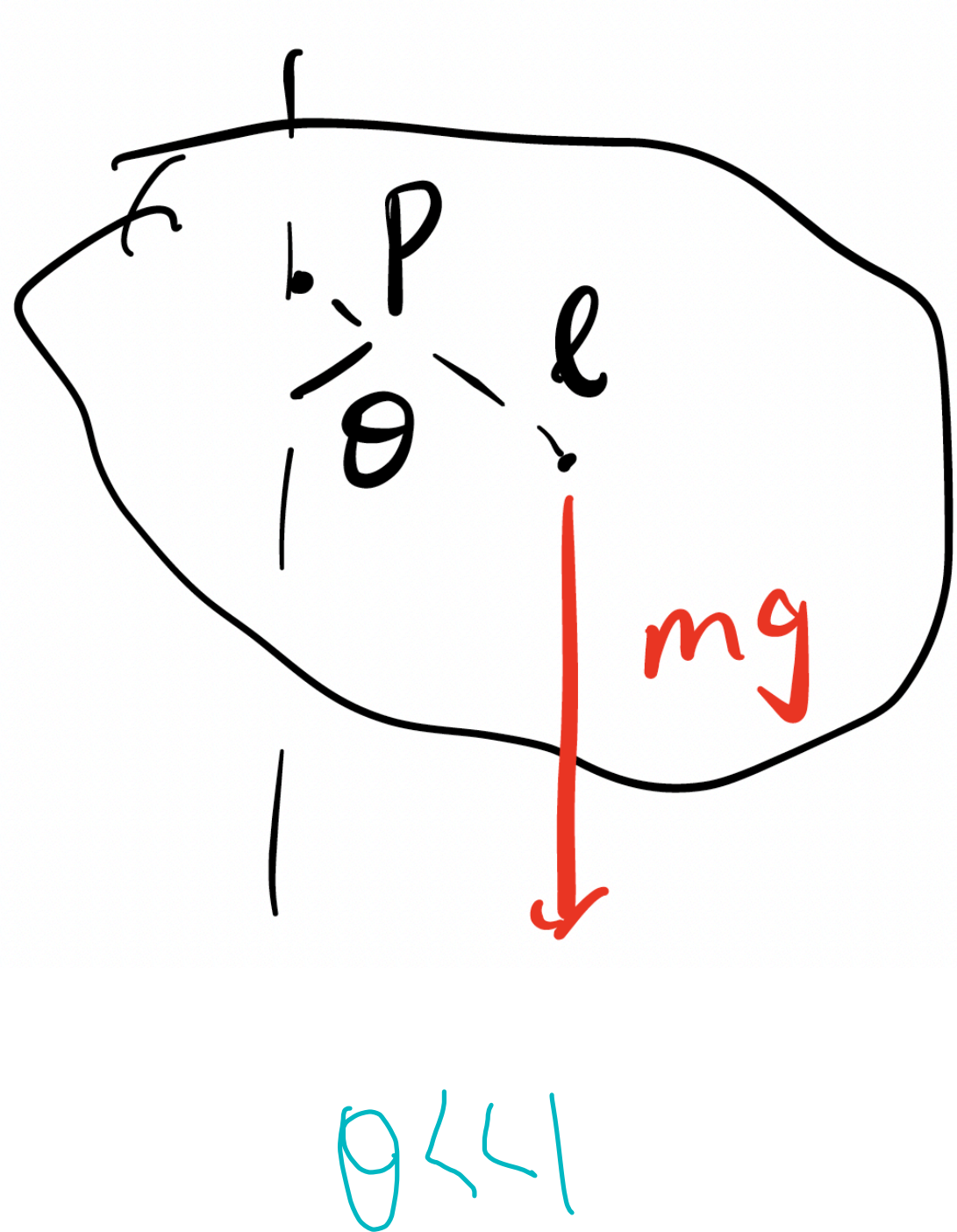
$$\omega = \sqrt{\frac{g}{4R}}$$

- (g) Find the angular frequency ω for the bead oscillating on the cycloidal wire. **(1 pt)**

You do not have to solve the equation of motion. Instead, you can use an analogy with the spring-mass system, which has the following equation of motion and angular frequency

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0, \quad \text{and} \quad \omega = \sqrt{\frac{k}{m}}.$$

Physical Pendulum



$$\tau = I \alpha$$

$$= I \frac{d^2\theta}{dt^2}$$

$$\tau = (\vec{r}) (\vec{F}) \sin \theta$$

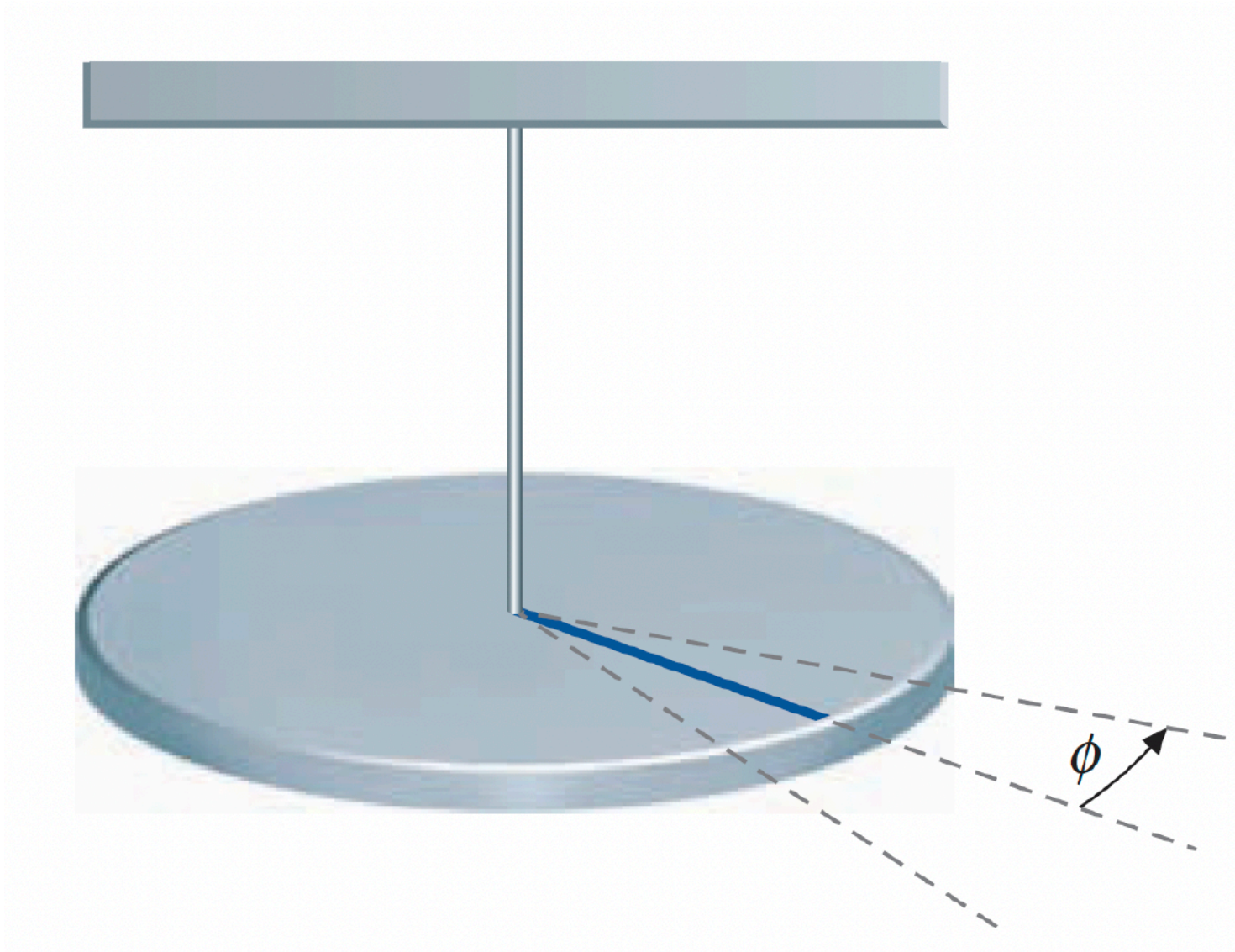
$$= -mg \cdot l \cdot \sin \theta$$

$$I \frac{d^2\theta}{dt^2} + mgl \cdot \theta = 0$$

$$\boxed{\frac{d^2\theta}{dt^2} + \frac{mgl}{I} \theta = 0}$$

EOM

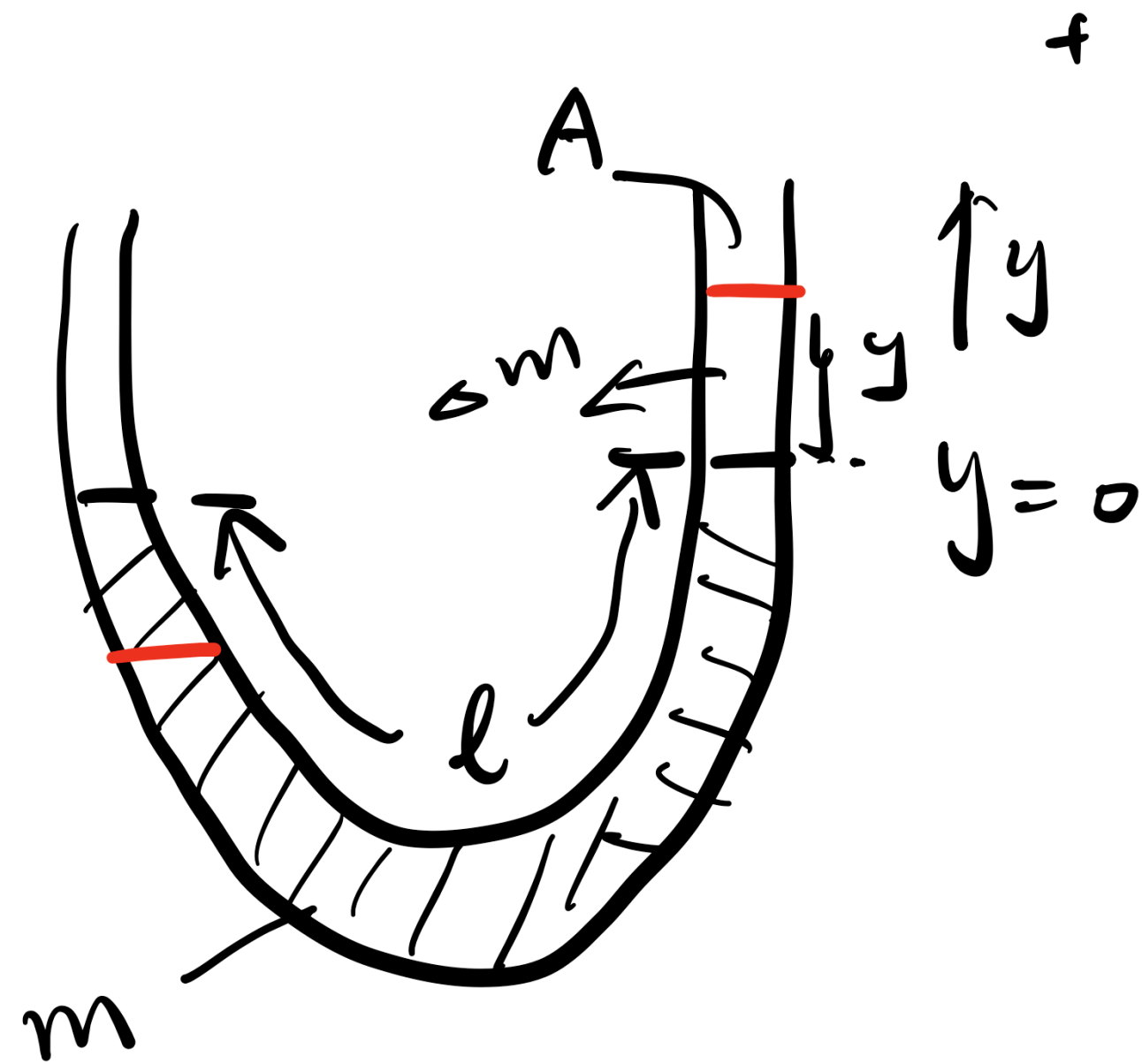
Torsional Pendulum



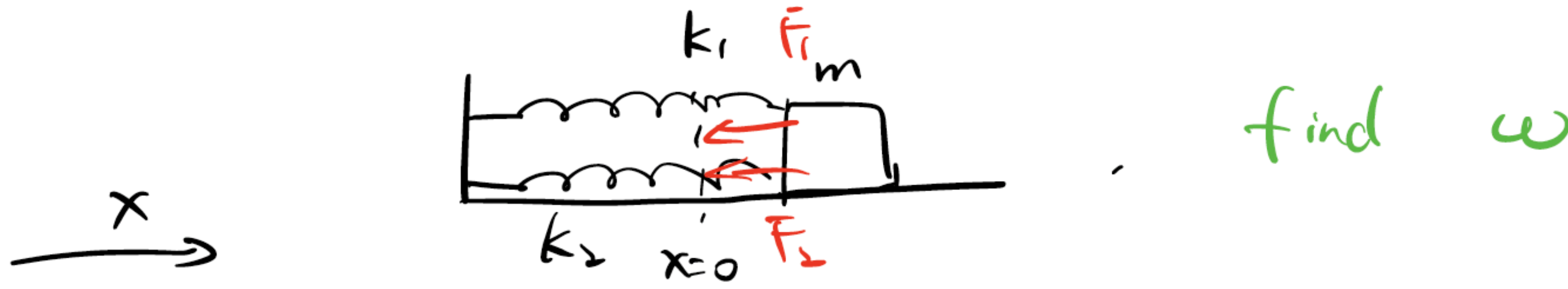
Energy of Oscillation

- $E = \frac{1}{2}kA^2$

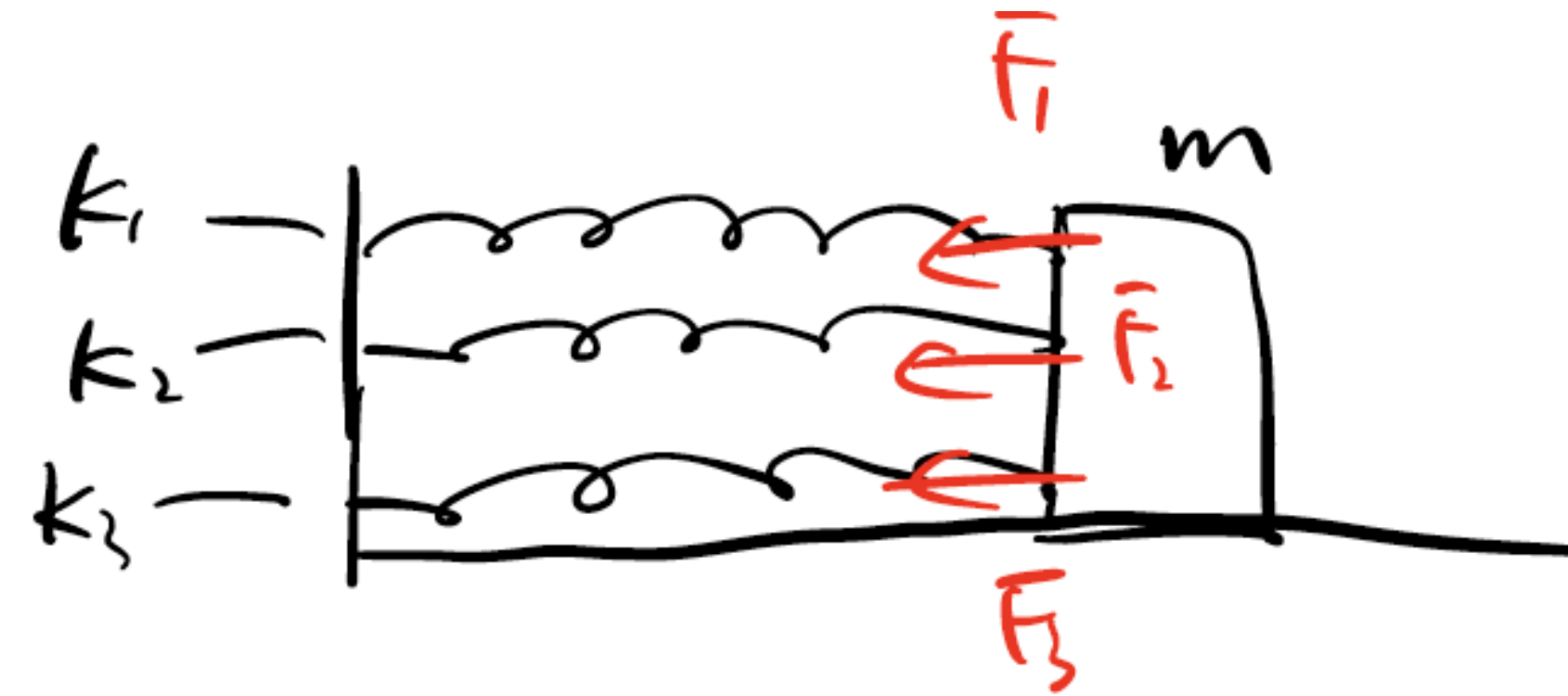
Fluid & Oscillation Revist



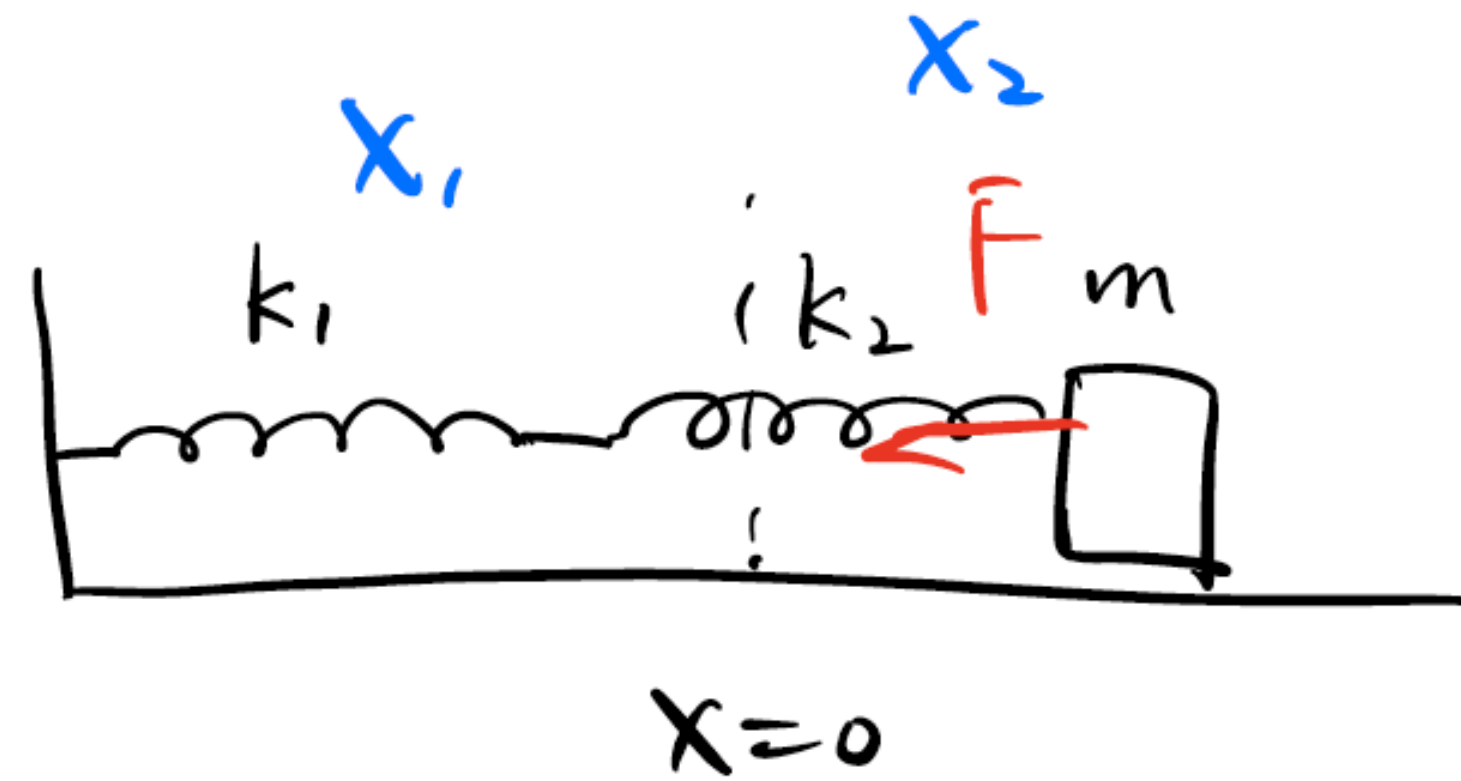
Springs in Parallel



Springs in Parallel



Springs in Series



Springs in Series

